

Dinesh Kothandaraman

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 Github |  Portfolio

Summary

Senior AI/ ML Engineer with 7+ years of hands-on experience designing, experimenting with, and deploying ML and generative AI systems. Strong background in LLMs, agent-based architectures, retrieval-augmented systems, and experimental evaluation. Proven ability to independently design research experiments, fine-tune models, build end-to-end prototypes, and translate research ideas into functional demos and production-ready solutions. Comfortable operating in fast-paced startup environments with minimal guidance.

Technical Skills

AI/ML & Gen AI: LLMs (OpenAI, Claude, Gemini), RAG, Agentic AI, LangChain, LangGraph, MCP workflows, Prompt Engineering, Supervised Fine-Tuning, NLP, Transformers (T5, BART)

Machine Learning & Research: PyTorch, TensorFlow, Scikit-learn, Model Fine-Tuning, Experimental Design, Metrics & Benchmarking

MLOps & DevOps: Docker, Kubernetes, CI/CD (GitHub Actions), MLflow, DVC, Airflow, Kubeflow

Data & Engineering: Apache Spark, PySpark, SQL, Data Pipelines, ETL

Programming & APIs: Python, JavaScript/TypeScript, FastAPI, Flask, REST APIs

Cloud & Platforms: AWS, Azure, GCP, Vertex AI

Databases: PostgreSQL, Snowflake, Vector Databases

Experience

Senior AI Data Engineer | Photon Infotech, Texas-United States| Apr 2026 – Present

- Designed and developed a multi-agent QA automation platform using LangGraph, JSON-RPC 2.0, and schema-driven workflows to orchestrate four AI-powered microservices for endpoint filtering, test intent generation, feature creation, and coverage governance.
- Built an end-to-end API test generation pipeline that analyzes Jira user stories and OpenAPI specifications, leveraging OpenAI/Azure OpenAI models to generate structured test intents and executable Behave (Gherkin) test suites for automated regression testing.
- Developed a full-stack solution with FastAPI, PostgreSQL, and React (TypeScript, Vite, Tailwind CSS), providing workflow orchestration, execution monitoring, coverage analytics, and reporting dashboards.
- Implemented LLM governance and security controls including schema validation, structured output enforcement, prompt safety mechanisms, input sanitization, and role-based API access controls to improve reliability, security, and compliance of AI-generated test artifacts.

Machine Learning Engineer | Sea 48 LLC, Connecticut-United States| Sep 2025 – Apr 2026

- Built a financial analytics dashboard to visualize forecasted occupancy rates and profit & loss metrics, enabling data-driven operational and financial planning.
- Developed time-series forecasting models on occupancy and revenue data to analyze trends and support forecasting accuracy.
- Designed and implemented a conversational chatbot using LangChain with multi-agent routing, allowing users to query financial data and receive contextual, role-aware responses.
- Exposed forecasting, analytics, and chatbot capabilities through scalable REST APIs using FastAPI,

supporting real-time queries and dashboard integration.

- Implemented data privacy and governance controls for sensitive financial and banking information, including secure data access and safeguards for handling financial data.

Agentic AI Developer | Kamali Inc, Remote-United States| Apr 2025 – Jun 2025

- Designed and executed applied research experiments on LLM-based retrieval, agent routing, and multi-step reasoning.
- Built end-to-end prototypes including data preparation, embedding strategies, agent orchestration, and evaluation.
- Experimented with agentic architectures (tool-calling, memory, intent routing) to improve reasoning accuracy and task completion.
- Designed evaluation frameworks to measure retrieval quality, response accuracy, and agent behavior.
- Translated research findings into working demos and production integrations, collaborating closely with engineers.
- Iterated rapidly on ideas, validating hypotheses through hands-on experimentation.

Machine Learning Engineer | Cognizant Technology Solutions, India | Jan 2021 – July 2023

- Designed and deployed enterprise-scale ML platforms for underwriting and risk assessment in highly regulated environments.
- Built distributed data pipelines using PySpark to process TB-scale structured and semi-structured datasets.
- Implemented model deployment, monitoring, and versioning to support multiple production systems.
- Ensured solutions met governance, auditability, and explainability standards, collaborating with risk and compliance teams.
- Delivered APIs and services integrated into downstream analytics and decision systems.

Machine Learning Engineer | Jubilee Information Technology Pvt Ltd, India | July 2017 – Dec 2020

- Developed ML-based product recommendation system to suggest personalized insurance plans based on demographic, behavioral, and claim history data, improving upsell conversions by 20%.
- Built customer segmentation pipelines using K-Means and hierarchical clustering to group policyholders by health risk, lifestyle indicators, and purchasing behavior.
- Processed large-scale policyholder and claim data using PySpark, performing feature engineering, outlier detection, and aggregation at household and individual levels.
- Automated daily batch scoring workflows using PySpark on AWS EMR, pushing predictions to downstream CRM systems via Kafka and PostgreSQL.

Publication & Certifications

Hybrid CNN-LSTM Model for Jail Activity Recognition, Oct 2024

Published in 2024 IEEE 15th Annual UEMCON, [Link](#)

Oracle Cloud Infrastructure 2025 Certified Data Science Professional, Oct 2025, [Link](#)

Projects

Document Based QA System (Conversational AI Chatbot)

- Developed a chatbot using Python and Retrieval-Augmented Generation (RAG) with large language models (LLMs) to answer questions based on document content, leveraging vector embeddings for efficient context retrieval and intent recognition.
- Orchestrated data processing and model inference using Apache Airflow to automate pipeline

execution, ensuring seamless integration of document ingestion and question-answering tasks.

- Implemented CI/CD workflows with GitHub Actions to automate model updates and monitoring, achieving robust production-grade deployment for a conversational agent.

Remote Patient Care with AWS and IoT (Healthcare Monitoring System)

- Designed a scalable health monitoring system integrating IoT devices with AWS tools, demonstrating experience relevant to the healthcare domain.
- Utilized AWS Glue for processing health data and Python for anomaly detection in real-time metrics, supporting proactive patient care.
- Developed dashboards in AWS QuickSight to visualize and monitor health indicators, providing actionable insights for healthcare providers.

Text Summarization - Natural Language Processing

- Developed three text summarization models: an LSTM with attention mechanism and two pre-trained transformer models (T5 and BART), focusing on both abstractive and extractive techniques.
- Fine-tuned T5 and BART models on domain-specific datasets, leveraging their encoder-decoder architecture for high-quality summarization with improved ROUGE and BLEU scores.
- Implemented an LSTM with attention to enhance focus on relevant text segments, visualizing attention weights for interpretability and improving model performance on custom datasets.

Car Logo Detection - Deep Learning

- Developed a car logo detection model using PyTorch and transfer learning, leveraging pre-trained CNN architectures (ResNet, VGG) to achieve high accuracy on custom datasets.
- Implemented transfer learning by fine-tuning a pre-trained model on a labeled car logo dataset, reducing training time and improving generalization.
- Integrated ClearML for streamlined experiment tracking, performance monitoring, and efficient model versioning during the development lifecycle.
- Optimized model performance through data augmentation, learning rate scheduling, and loss function tuning, achieving significant improvement in precision and recall metrics.

Jail Activity Recognition Using CNN and LSTM

- Built a real-time activity recognition model combining CNN and LSTM architectures.
- Processed and analyzed time-series data, achieving high model accuracy.

Gender and Age Detection

- Developed a Python-based deep learning project to detect faces and classify age and gender from images.
- Leveraged pre-trained models to achieve high accuracy and real-time performance.
- Utilized advanced computer vision techniques for optimized analysis.

Education

MS, University of New Haven, Data Science | Aug 2023 – May 2025 | GPA - 3.89/4.0

BE, Chennai Institute of Technology, Mechatronics Engineering | June 2013 – April 2017 | GPA - 7.35/10